

Weekly Progress Report - Week 2

Improve performance of existing online trackers for small object detection [UAV Videos]

Group: The Convolutioners

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Outline of Performed Tasks:

- Implemented a modular framework to replace deep ReID embeddings with handcrafted classical features
- Analyzed tracker-specific implementations (ByteTrack, DeepSORT, BoT-SORT)

Dataset Update for MOT Experiments

The dataset used for the project has been changed to the **VisDrone dataset**. This change was made because the previously considered **AU Drone dataset** primarily contains individual images rather than continuous video sequences.

Since multi-object tracking (MOT) algorithms require sequential frames to track object identities over time, a dataset with video data is necessary. The VisDrone dataset provides UAV-captured video sequences with annotations suitable for tracking tasks, making it more appropriate for evaluating MOT algorithms.

BoxMOT Framework

We use the BoxMOT framework as the base system for implementing and testing different multi-object tracking algorithms. BoxMOT provides a modular structure that connects object detectors (e.g., YOLO) with tracking algorithms and allows easy integration of different appearance models. This flexibility makes it suitable for our work, where we aim to experiment with replacing deep ReID embeddings with classical handcrafted features.

Reference Materials Studied

1. ByteTrack

Studied the ByteTrack tracking method, which associates detections using motion prediction and considers both high-confidence and low-confidence detections to improve tracking recall.

2. DeepSORT

Reviewed DeepSORT, which extends the SORT tracker by adding appearance-based ReID embeddings to improve identity association during occlusions and long-term tracking.

3. BoT-SORT

Explored BoT-SORT, which improves upon ByteTrack by incorporating stronger ReID features and additional refinements in the data association process.

Tentative List of Tasks for Next Week:

1. Implement and evaluate a classical computer vision feature extractor (e.g., HOG or color histogram) for object appearance matching in the MOT pipeline